Roll No. ....

TCE–603

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

WATER RESOURCES ENGINEERING–II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) State the factors that are to be considered while selecting suitable site for a given dam. (CO1)

OR

(b) Determine the wave force and free board given : (CO1)

Wind velocity = 100 km/hr

Fetch = 34 km

Average depth of reservoir along fetch = 18 m

2. (a) Explain the middle-third rule criteria for the stability analysis of gravity dam. (CO1)

P. T. O.

OR

(b) Design the practical profile of a gravity dam of stone masonry considering full uplift, from given the following data : (CO1)

- R.L. of base of the dam = 1550 m
- R. L. of F. R.L. = 1590.50 m
- Specific gravity of the masonry = 2.40
- Height of waves = 2.0 m
- Safe compressive stress for masonry = 1200 kN/m²

3. (a) Discuss briefly the various causes of failure of earthen dams. (CO2)

OR

(b) Determine the pheratic line for an earthen dam homogeneous in section from the following data : (CO2)

Height of dam = 20 m

Free board = 2.0 m

Coefficient of permeability of material of the dam = 4×10^{-4} cm/sec

Top width = 4 m

Upstream slope = 3 : 1

Downstream slope = 3 : 1

An horizontal filter provided extends to 30 m from the toe point of the dam.

And also find discharge per meter length of the earthen dam.

4. (a) Explain the various methods of controlling seepage through an earthen dam. (CO2)

OR

(b) Determine the factor of safety of downstream slope of an earthen dam(homogeneous section) drawn to a scale of 1 cm = 5 m : (CO2)

Length of slip circle arc	=	15 cm
Total area of N-rectangle	=	16.5 cm²
Total area of T-rectangle	=	7.0 cm²
Total area of U-rectangle	=	5 cm²
Angle φ	=	260
Cohesion c	=	20 kN/m²
Specific weight of soil	=	18 kN/m³

5. (a) Explain the thin cylinder theory of design of an arch dam. What are the limitations of this theory ? (CO1, CO2)

OR

(b) A 10 m high concrete dam having trapezoidal section has top width 1 m and bottom width 8 m. The face of the dam exposed to water has a slope 0.1 H : 1 V. On the reservoir side water stand up to the top and no tail water. Assuming unit weight of conc. is 24 kN/m³, and taking in to account only the weight of the dam, water pressure and uplift pressure, calculate the factor of safety against overturning. (CO1, CO2)

Roll No. ....

TCE–602

B. TECH. (CE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

REINFORCED CEMENT CONCRETE—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

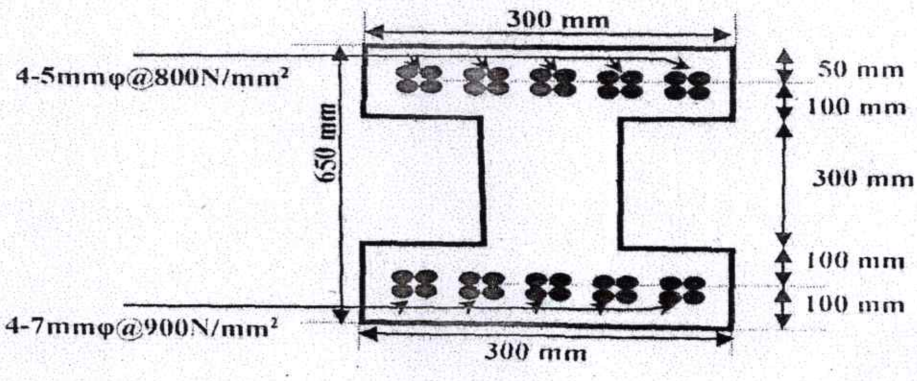
(ii) Each sub-question carries 10 marks.

1. (a) Explain the various methods of analyzing a prestressed member section. (CO1)

OR

(b) Explain the pre-tensioning methods mentioning at least *five* different application areas of pre-tensioning. (CO1)

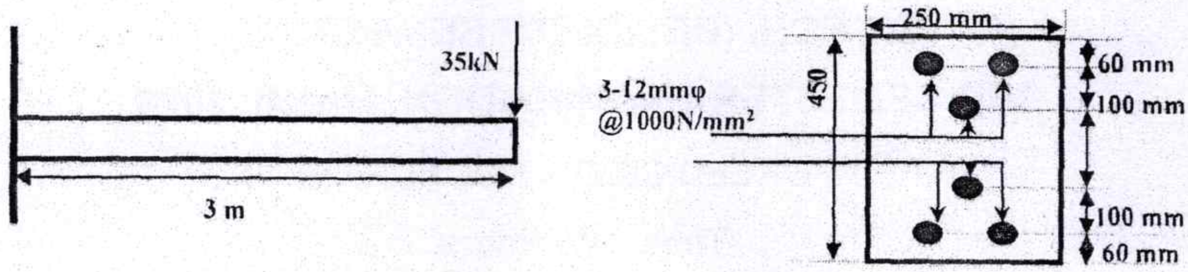
2. (a) For the given prestressed section, find the cable force is eccentricity. (CO1)



P. T. O.

OR

(b) Determine the extreme fiber stresses at L/7 span section from support for the given beam. (CO1)



3. (a) Derive the formulas for evaluating the prestress loss for various cases. (CO1)

OR

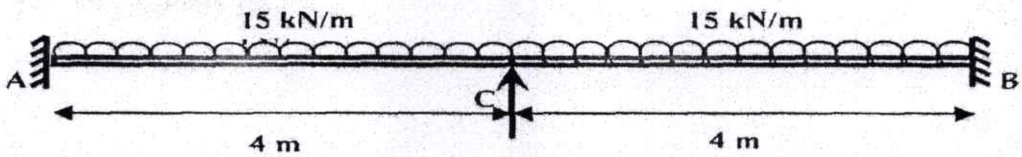
(b) Elaborate the various recommendations and assumptions by IS 456 : 200 code for moment redistribution in statically indeterminate structures. (CO2)

4. (a) Determine and draw the design bending moment diagram of the 6 meter long RC fixed beam clamped at both ends and carrying ultimate uniformly distributed load of 15 kN/m with full redistribution of 30 per cent as per IS 456. (CO2)

OR

(b) Explain the concept of moment redistribution and evaluate the shift of point of contra-flexure in a propped cantilever beam carrying a udl of 'w' over its entire span. (CO2)

5. (a) Draw the bending moment diagram of the two span continuous beam with a dead load of 4 kN/m. Develop the bending moment envelop applying 30% moment redistribution for the beam. (CO2)



OR

(b) Derive expressions for shear force, bending moment and twisting moment for one span of a complete circular beam simply supported on equally spaced supports, and subjected to a uniformly distributed load.

(CO2)

60

Roll No. ....

TCE–604

B. TECH. (CIVIL ENGINEERING)

(SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

QUANTITY ESTIMATION AND COSTING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain estimate and principle of estimate. Also explain its purpose and importance of estimates. (CO1)

OR

(b) Explain the report of an estimate. Support your answer with a brief description. (CO1)

2. (a) Explain the units of measurements and its importance in civil engineering works. (CO1)

OR

(b) Explain different methods of taking out quantities of items of work in preparing estimate. (CO1)

P. T. O.

3. (a) Explain data required to prepare an estimate. (CO1)

OR

(b) Prepare a preliminary estimate of a four storied office building having total carpet area of 4500 sq. m for obtaining the administrative approval of the government, given the following data. It may be assumed that 30% of the built up area will be taken up by corridors, verandah, lavatories, staircase etc. (CO1)

Plinth area rate is ₹ 2,250 per sq. m.

Consider the following charges in preparing preliminary estimate :

(i) Extra for architectural treatment 0.5% of building cost.

(ii) Extra due to deeper foundation at site 2% of building cost.

(iii) Extra for internal electrical installation 15% of building cost.

(iv) Extra for water supply and sanitary installation 8% of building cost.

(v) Extra for other services 5% of building cost.

(vi) Contingencies – 3%

(vii) Supervision charges – 10 %

4. (a) Explain the purpose and importance of rate analysis. (CO2)

OR

(b) Explain rate analysis and its effect on construction works. (CO2)

5. (a) Calculate the quantity and amount of material and labour for random rubble stone masonry in cement mortar (1 : 5) in foundation and plinth. Assume suitable rates. (CO2)

OR

(b) Calculate the quantity for material and labour for reinforced concrete beam excluding centering and shuttering. Assume suitable data for rate analysis. (CO2)

60

Roll No. ....

TCE–605

B. TECH. (CIVIL ENGINEERING) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

TRANSPORTATION ENGINEERING–I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain first, second and third 20 year road plan. (CO1)

OR

(b) Explain briefly the modified classification of road system in India as per third 20 year road development plan, 1981-2001. (CO1)

2. (a) Write short notes on the following : (CO1)

(i) Traffic separators

(ii) Kerbs

(iii) Shoulders

(iv) Width of formation

OR

(b) The design speed of a road is 65 km/hr, the friction coefficient is 0.36 and reaction time of driver is 2.5 sec. Calculate the values of : (CO1)

(i) Headlight sight distance.

(ii) Intermediate sight distance required for the road.

P. T. O.

3. (a) Design the rate of super elevation for a horizontal highway curve of radius 500 m and speed 100 kmph. (CO2)

OR

(b) A valley curve is formed by a descending grade if 1 in 25. Meeting an ascending grade of 1 in 30. Design the length of valley curve to fulfill both comfort condition and head light sight distance requirement for design speed of 80 kmph. Assume allowable rate of change of centrifugal acceleration $C = 0.6$ m/sec^3. (CO1)

4. (a) What are the desirable properties of sub grade soil ? Enumerate the identification and classification tests of soil. (CO2)

OR

(b) Explain the principle of conducting LOS angles abrasion test. (CO2)

5. (a) Explain the uses of bitumen emulsion. How are they prepared ? (CO2)

OR

(b) Explain briefly Marshall method of bituminous mix design. (CO2)